Schrödinger factorization chain for a system displaying broken supersymmetry

Bennett Lamb,[1] Ramkrishna Sharma,[1] Luogen Xu,[2] and James K. Freericks[2]
[1)]*Department of Physics, University of Virginia, P.O. Box 400714, Charlottesville, VA 22904 USA*
[2)]*Department of Physics, Georgetown University, 37th and O Sts. NW, Washington, DC 20057 USA*

(*Electronic mail: james.freericks@georgetown.edu)

(Dated: June 10, 2026)

The factorization method was discovered by Schrödinger in 1940 and it provides a systematic approach to algebraically determine energy eigenstates. In the 1980s, this approach was generalized by Witten to supersymmetric quantum mechanics, which works with two partner Hamiltonians, that share all energy eigenvalues, except for the ground state of the first potential in the pair. Witten also described the situation of broken supersymmetry, where the Hamiltonians for the two partner potentials share all eigenvalues. The classic example of broken supersymmetry uses a quadratic superpotential. The two partner potentials are mirror images of each other, so the two Hamiltonians naturally share the same eigenvalues. In this work, we show that these "supersymmetry breaking potentials" can still be factorized into the conventional factorization chain of Schrödinger, because the factorization of a Hamiltonian is often not unique. We show how one can construct this factorization numerically, and explain the behavior that underlies supersymmetry breaking. This work clarifies a common misconception that supersymmetry breaking potentials have Hamiltonians that cannot be factorized into a factorization chain. Instead, supersymmetry breaking is associated with the superpotential, not the potential. Every well-behaved one-dimensional potential can have a factorization chain constructed for it and we provide a numerical way to do that.

I. INTRODUCTION

The factorization method is an approach to solving for energy eigenstates in quantum mechanics using algebraic methods rather than differential equations. It was invented by Schrödinger[1–4] in 1940 and reworked by Witten in the 1980s as supersymmetric quantum mechanics.[5] The factorization method seeks to factorize a Hamiltonian in the form $\hat{H} = \hat{p}^2/2M + V(\hat{x}) = \hat{A}^\dagger\hat{A} + E$, with the Schrödinger ladder operators given by

$$\hat{A} = \frac{1}{\sqrt{2M}}\left(\hat{p} - i\hbar k W(k'\hat{x})\right), \tag{1}$$

and its Hermitian conjugate. Here, M is the mass of the particle, $\hat{x}$ is the position operator, $\hat{p}$ is the momentum operator, $V(\hat{x})$ is the potential, E is the ground-state energy, W is the superpotential (which is real valued) and k and k' are numbers with dimensions of inverse length. The position and momentum operators satisfy the canonical commutation relation $[\hat{x},\hat{p}] = i\hbar$. In the factorization method, one constructs a factorization chain with a family of ladder operators indexed by an integer, such that the initial Hamiltonian corresponds to index 0 and the factorization chain satisfies

$$\hat{H}_n = \hat{A}_n^\dagger\hat{A}_n + E_n = \hat{A}_{n-1}\hat{A}_{n-1}^\dagger + E_{n-1}, \tag{2}$$

with the ground state of each auxiliary Hamiltonian for $n \geq 1$ satisfying the subsidiary condition $\hat{A}_n|\phi_n\rangle = 0$ and having energy E_n. The way the chain is constructed is we start from the original Hamiltonian and factorize it. Then we construct $\hat{H}_1 = \hat{A}_0\hat{A}_0^\dagger + E_0$ by using the Schrödinger ladder operators in the opposite order. This has a new potential $V_1(\hat{x})$ and we need to factorize that new Hamiltonian with the new Schrödinger ladder operators indexed by 1. The nth excited state of the original Hamiltonian is constructed from

$$|\psi_n\rangle = \frac{1}{\sqrt{(E_n-E_0)(E_n-E_1)\cdots(E_n-E_{n-1})}} \times \hat{A}_0^\dagger\hat{A}_1^\dagger\cdots\hat{A}_{n-1}^\dagger|\phi_n\rangle, \tag{3}$$

and has energy E_n. In general, we cannot analytically factorize all Hamiltonians—nevertheless all known exactly solvable Hamiltonians, with so-called shape-invariant potentials, can be algebraically factorized in this way.

Supersymmetric quantum mechanics, on the other hand, focuses on just a Hamiltonian and its partner, which are the two forms $\hat{H}_0 = \hat{A}^\dagger\hat{A}$ and $\hat{H}_1 = \hat{A}\hat{A}^\dagger$; in supersymmetric quantum mechanics, the potential of $\hat{H}_0$ is shifted via $V_0 \to V_0 - E_0$, so that the ground-state energy is zero. Then, the ground state of the first Hamiltonian satisfies $\hat{A}|\psi\rangle = 0$ with $E_0 = 0$. All excited energies of the first Hamiltonian are shared with the second Hamiltonian, but the second Hamiltonian's ground state lies at $E > 0$. Supersymmetry is said to be broken if there is no ground state of the first Hamiltonian with $E_0 = 0$. In other words, the ground state does not satisfy $\hat{A}|0\rangle = 0$, so that $E_0 \neq 0$. It is often the case then that the two Hamiltonians share all of their energy levels.

In cases where the Hamiltonian cannot be directly factorized algebraically, it still can be factorized formally. If we know the ground state wavefunction of $\hat{H}_0$, which is given by $\psi_0(x) = \langle x|\psi_0\rangle$, then we can always choose $kW(k'\hat{x}) = -\psi_0'(\hat{x})/\psi_0(\hat{x})$. Note that we construct the operator $\psi(\hat{x})$ by finding the functional form of $\psi(x)$ and formally replacing $x \to \hat{x}$ and similarly for its derivative. A direct calculation

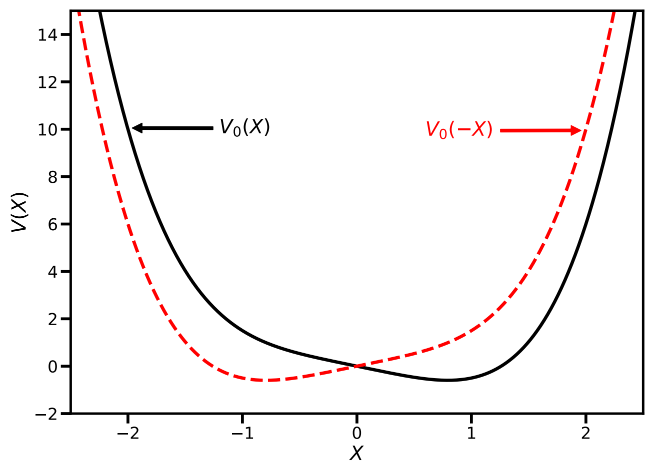

Figure 1. Original potential $V_0(X)$ (black solid) and its supersymmetric partner (red dashed), which is its mirror image $V_0(-X)$. This is the common example used to illustrate supersymmetry breaking when it is described by an even quadratic superpotential. The energy unit is $\hbar^2k^2/M$ and the dimensionless position is $X = kx$.

yields

$$\begin{aligned}\hat{A}_0^\dagger\hat{A}_0 &= \frac{\hat{p}^2}{2M} - \frac{i\hbar k}{2M}[\hat{p}, W(k'\hat{x})] + \frac{\hbar^2k^2}{2M}W^2(k'\hat{x}) \\ &= \frac{\hat{p}^2}{2M} + \frac{i\hbar}{2M}\left[\hat{p}, \frac{\psi_0'(\hat{x})}{\psi_0(\hat{x})}\right] + \frac{\hbar^2}{2M}\left(\frac{\psi_0'(\hat{x})}{\psi_0(\hat{x})}\right)^2 \\ &= \frac{\hat{p}^2}{2M} + \frac{\hbar^2}{2M}\frac{\psi_0''(\hat{x})}{\psi_0(\hat{x})} = \frac{\hat{p}^2}{2M} + V(\hat{x}) - E_0, \end{aligned} \tag{4}$$

because $\psi_0(\hat{x})$ satisfies the Schrödinger equation given by $-\hbar^2\psi''(\hat{x})/2M + V(\hat{x})\psi(\hat{x}) = E_0\psi(\hat{x})$. So, in principle, any potential can be factorized and one can construct a factorization chain for the system, if desired. This construction follows from the well-known equivalence of the Schrödinger equation with the Riccati equation.[6] Note that because the ground-state wavefunction has no nodes, the superpotential is nonsingular. In addition, because the ground-state wavefunction vanishes as $x \to \pm\infty$, we must have the superpotential be positive for $x \to \infty$ and negative for $x \to -\infty$ for a normalizable wavefunction. Furthermore, this approach provides a nice connection between the superpotential and the wavefunction, since we have

$$\psi(x) = \psi(x_0)\exp\left(-\int_{x_0}^{x} dx' kW(k'x')\right) \tag{5}$$

for an appropriately chosen x_0.

When supersymmetric quantum mechanics is discussed in instruction (see for example Ch. 3 of Cooper[7] *et al.* or p. 508 of Zwiebach[8]), it is common to use an even superpotential (the simplest being $kW(k'\hat{x}) = k^3\hat{x}^2$) which breaks supersymmetry. The original potential becomes $V_0(\hat{x}) = \hbar^2k^6\hat{x}^4/2M - \hbar^2k^3\hat{x}/M$ and the partner potential is $V_1(\hat{x}) = \hbar^2k^6\hat{x}^4/2M + \hbar^2k^3\hat{x}/M$. A schematic plot is shown in Fig. 1. Because of the broken supersymmetry, we cannot find the ground state by exponentiating the integral of the superpotential, and it appears as if one cannot construct a factorization chain for these potentials that illustrate supersymmetry breaking. The traditional construction fails because the integral of the superpotential in Eq. (5) does not create a *normalizable* wavefunction when we have broken supersymmetry. Nevertheless, this construction does immediately tell us why the energy levels are degenerate, because we have $V_1(-\hat{x}) = V_0(\hat{x})$ and energy eigenvalues do not change if we transform $\hat{x} \to -\hat{x}$ in the Schrödinger equation. These broken supersymmetry rules hold for any superpotential that is an even function, because it cannot solve the normalizability requirements of changing sign of the superpotential and the potential generated by it and its superpartner are always mirror images.

Because the ground state for $\hat{H}_0$ is square integrable and has no nodes, even for a supersymmetry breaking situation it must also have a traditional Schrödinger factorization chain. In this work we show how this is so by constructing the first few links of the factorization chain numerically. This work also dispels the misconception that potentials can be supersymmetry breaking—because the factorization is not unique, it is the potential *and the specific choice of the superpotential* that determines whether a system is supersymmetry breaking or not. For example, all potentials generated by even superpotentials must be supersymmetry breaking.

As we mentioned above, the factorization of a Hamiltonian is often not unique. There are numerous examples for extra factorizations that we will summarize below. First, we will discuss why any potential in one-dimension that has at least one bound state will have a standard factorization chain. The reason why is that we can always form a factorization chain based on the ground-state wavefunctions of each of the auxiliary Hamiltonians. It does require us to be able to diagonalize the Hamiltonians, but it then reveals the potentials for the isoenergetic Hamiltonians (except for one state). The strategy is as follows: (i) for the first Hamiltonian, we determine the ground state $\psi_0(x)$ and form $kW_0(k'\hat{x}) = -\psi_0'(\hat{x})/\psi_0(\hat{x})$; (ii) using the superpotential and the ground-state eigenvalue E_0, we next compute $\hat{H}_1 = \hat{A}_0\hat{A}_0^\dagger + E_0$, which has a different potential, which we call $V_1(\hat{x})$; (iii) we diagonalize $\hat{H}_1$ to find its ground state $\phi_1(x)$ and we use it to find the next superpotential $kW_1(k'\hat{x}) = -\phi_1'(\hat{x})/\phi_1(\hat{x})$; (iv) we then form $\hat{H}_2 = \hat{A}_1\hat{A}_1^\dagger + E_1$, which determines the next potential $V_2(\hat{x})$; and (v) we continue in this fashion computing the next ground-state wavefunction and energy and then determining the next potential in the chain. This approach will always work because the ground state of the original and all auxiliary Hamiltonians has no nodes, so the superpotential is always well defined and never singular.

In this procedure, the potentials that are formed for the next link of the chain take the form

$$V_{n+1}(\hat{x}) = \frac{\hbar^2}{M}\left(\frac{\phi_n'(\hat{x})}{\phi_n(\hat{x})}\right)^2 - V_n(\hat{x}) + 2E_n. \tag{6}$$

As long as these potentials are bounded from below, the chain can be extended another link. Whether this stops after a finite number of links (and thereby a finite number of bound states) or continues for an infinite number of bound states varies for different potentials. If the potential does not support at least two bound states, the chain will end as well. This factorization chain is one that always exists.

There is one general case where a second factorization always exists—this is when the potential is an even function of x and has at least two bound states. In this case, we can always form a second factorization chain starting from $kW(k'\hat{x}) = -\psi_1'(\hat{x})/\psi_1(\hat{x})$—the logarithmic derivative of the first excited state. This superpotential is singular at the origin and it only produces the odd eigenstates of the original Hamiltonian. The key for this is that all odd functions vanish at the origin, so all of the superpotentials in the factorization chain will remain singular at the origin as well. A nice example of this is a second factorization of the simple harmonic oscillator, which includes a superpotential of the form $\alpha/\hat{x}+\beta\hat{x}$. By adjusting the constants, we can form the simple harmonic oscillator potential, but with a superpotential that is singular at the origin; this factorization is identical to the s-wave factorization of an isotropic harmonic oscillator in three dimensions.

Otherwise, there are many other factorizations possible, since we always have $kW(k'\hat{x}) = -\psi_n'(\hat{x})/\psi_n(\hat{x})$ as a singular superpotential that can be used for a factorization. These factorizations usually cannot be extended to factorization chains, because the singularities separate the system into domains where the wavefunction must vanish. But there is no reason for the potentials in each domain to support the same energy eigenvalues across all of these domains.[10] Hence, forming the first link in the factorization chain usually creates a piecewise potential that does not support an energy eigenstate with the same energy across the different domains. Hence, these factorizations cannot be extended to valid factorization chains.

Finally, if we have a potential constructed from an even superpotential, then the potential will not be even, but the potential for $\hat{A}^\dagger\hat{A}$ and for $\hat{A}\hat{A}^\dagger$ will be mirror reflections of each other—$V_0(\hat{x}) = V_1(-\hat{x})$, which can be easily seen from the relationship between the superpotential and the potential. This is the case for the form of broken supersymmetry that we consider here. Indeed, an even superpotential never allows for a normalizable wavefunction when we solve the subsidiary condition, so it always corresponds to broken supersymmetry.

Of course, the only way to do this general factorization is to proceed numerically, so we discuss how we do this next.

II. NUMERICS

There are numerous ways to numerically solve for the eigenvalues and eigenvectors of the Schrödinger equation. All of them are approximate in one form or another because they do not fully encompass the infinite domain. One of the most popular methods is the shooting method, which starts at a large negative value to the left, setting the wavefunction to zero and having a unit derivative. Then the differential equation is used to translate the wavefunction to the right using a guess for the energy eigenvalue. When one reaches the far right, one of three options has occurred—(i) the wavefunction is diverging towards $+\infty$; (ii) the wavefunction is diverging towards $-\infty$; or (iii) the wavefunction remains close to zero. Only when the third case occurs have we found the energy eigenvalue and the wavefunction. If either the first or second occurs, the energy is adjusted and the shooting method is run again. This approach can be quite laborious to undertake, especially if no good estimates exist for the energy, but with trial and error and repetition, it will be successful with care taken to assure convergence. The only errors that occur are the accumulated errors from the discretization of the differential equation. One only obtains the wavefunctions and energy eigenvalues one at a time this way. The node theorem can be employed to determine which energy level has been found. It can become increasingly more difficult to determine higher energy eigenstates this way due to accumulated error requiring smaller and smaller spatial steps, but with care this approach can work for many cases.

Another strategy is to map the problem to a matrix equation by discretely gridding a finite position domain outside of which the wavefunction is expected to be negligibly small. Then we use discrete approximations to the Hamiltonian on the grid to construct an approximation of $\hat{H}$ in a discrete matrix format. Next one simply diagonalizes the matrix to determine its eigenvalues and eigenvectors. Since this approach requires the wavefunctions to vanish at the endpoints, it can only produce accurate results for wavefunctions that are negligible near the endpoints. Hence, once the energy is large enough, the results will become inaccurate because the wavefunction will be nonnegligible in the vicinity of $\pm x_{\max}$. There are a number of choices for how to construct the discrete representation of the Hamiltonian. The most common is to use a discrete representation of the second derivative with the grid point and its two neighbors to the right and left. An alternative approach is the discrete variable representation[9] (DVR) method, which uses a complete set of states truncated to low energies to compute the kinetic energy approximation. More specifically, it uses sinc functions localized at each lattice site.

In this work, we use the DVR method to perform these calculations. The explicit form we took is arrived at in the following way. We choose a cutoff value for the position grid $-x_{\max} \le x \le x_{\max}$ and a grid spacing Δx, which then determines the number of points in the grid. The discrete form of the Hamiltonian becomes

$$\begin{aligned} H_{nn} &= \frac{\pi^2\hbar^2}{6M(\Delta x)^2} + V(n\Delta x) \text{ and} \\ H_{mn} &= \frac{\hbar^2(-1)^{m-n}}{M(\Delta x)^2(m-n)^2} \text{ for } m \ne n. \end{aligned} \tag{7}$$

One typically must keep N small enough that the matrix diagonalizations can be completed in a reasonable time frame (in our work we keep the maximum matrix size below two thousand). The spatial cutoffs must be chosen large enough that the wavefunction vanishes at the endpoints to a good approximation.

The procedure for how to solve for the factorization chain is now straightforward. We use ϕ_n to denote the ground state wavefunction of $\hat{H}_n$ (so that $\psi_0 = \phi_0$), and ψ_n to denote the energy eigenstates of $\hat{H}_0$, the original Hamiltonian. Then, after multiplying the factorization chain equations from the left by $\langle x|$, we find that we must implement the following strategy.

The initial potential $V_0(x)$ is given to us. Using this potential, we solve for the ground state wavefunction and energy, denoted $\psi_0(x) = \phi_0(x)$ and E_0, respectively. Then the next potential is constructed from the factorization chain procedure, which yields $V_1(x) = -V_0(x) + 2E_0 + \hbar^2\big(\phi_0'(x)/\phi_0(x)\big)^2/M$. The derivatives of the ground-state wavefunction of the original Hamiltonian ϕ_0 must be computed numerically. The procedure then continues. Once the nth potential is found, we use the DVR method to compute its ground state $\phi_n(x)$ and then compute the new potential via $V_{n+1}(x) = -V_n(x) + 2E_n + \hbar^2\big(\phi_n'(x)/\phi_n(x)\big)^2/M$. One can continue doing this procedure indefinitely, but eventually the errors from the discretization will start to build up and cause problems. Here, we carry this procedure out to $n = 3$. From a numerical standpoint, once the wavefunctions become too small in size, numerically determining the logarithmic derivative becomes problematic. We describe in the next section how to handle this via an extrapolation procedure.

Next, we compute the wavefunctions for the excited states of the original Hamiltonian. The calculations are aided by the subsidiary condition, which converts $\hat{A}_n|\phi_n\rangle = 0$, given by

$$\left(\hat{p} + i\hbar\frac{\phi_n'(\hat{x})}{\phi_n(\hat{x})}\right)|\phi_n\rangle = 0, \tag{8}$$

into

$$\hat{p}|\phi_n\rangle = -i\hbar\frac{\phi_n'(\hat{x})}{\phi_n(\hat{x})}|\phi_n\rangle. \tag{9}$$

So, the first excited state satisfies

$$\begin{aligned}|\psi_1\rangle &= \frac{1}{\sqrt{(E_1 - E_0)}}\hat{A}_0^\dagger|\phi_1\rangle \\ &= \frac{1}{\sqrt{2M(E_1 - E_0)}}\left(\hat{p} - i\hbar\frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right)|\phi_1\rangle \\ &= \frac{-i\hbar}{\sqrt{2M(E_1 - E_0)}}\left(\frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})} + \frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right)|\phi_1\rangle. \end{aligned} \tag{10}$$

We then multiply by the bra $\langle x|$ from the left and let $\hat{x}$ act on the bra and be replaced by x to determine the wavefunction for the first excited state. Note that this wavefunction is already normalized. This gives

$$\psi_1(x) = \frac{-i\hbar}{\sqrt{2M(E_1 - E_0)}}\left(\phi_1'(x) + \frac{\phi_1(x)\phi_0'(x)}{\phi_0(x)}\right) \tag{11}$$

as the normalized first excited state of the original Hamiltonian; one can remove the overall global phase factor to make the wavefunction real valued, if desired. We can proceed in a similar fashion for the second and third excited states, and the results of these calculations are summarized in the appendix.

When we calculate higher excited states, we find terms that depend on higher derivatives of the auxiliary ground state wavefunctions $\phi_n(x)$. Numerically calculating high-order derivatives is a challenging task. But we can use the Schrödinger equation to replace second-order derivatives of the wavefunction by the potential minus the energy multiplying the wavefunction because

$$\phi_n''(x) = \frac{2M}{\hbar^2}\big(V_n(x) - E_n\big)\phi_n(x). \tag{12}$$

The calculation of the right hand side is much more stable than that of the left hand side. So, in the process of computing the higher-energy eigenstates, we need to remove higher-order derivatives using this identity to keep the highest numerical derivative that we must calculate being only the first derivative. This helps stabilize the calculations. Final results for the potentials and for the excited state wavefunctions are given in the appendix as well.

III. RESULTS

A. Analytical Verification with Shape Invariant Potentials

Before evaluating the numerical performance of the factorization chain on potentials with broken supersymmetry, it is instructive to apply our derived formulas to two exactly solvable shape-invariant systems. This provides a rigorous analytical check of the normalization constants and the derivative expansions up to the third excited state.

The first case is the simple harmonic oscillator (SHO), defined by $V_0(x) = \frac{1}{2}M\omega^2x^2$, which has the well-known energy spectrum $E_n = (n + \frac{1}{2})\hbar\omega$. The normalized ground-state wavefunction of the initial Hamiltonian is

$$\phi_0(x) = \left(\frac{M\omega}{\pi\hbar}\right)^{1/4}\exp\left(-\frac{M\omega}{2\hbar}x^2\right). \tag{13}$$

The logarithmic derivative of this ground state is a linear function in x. For simplicity in the following expressions, we define the constant $c = M\omega/\hbar$, yielding

$$\frac{\phi_0'(x)}{\phi_0(x)} = -cx. \tag{14}$$

To construct the factorization chain, we evaluate the first auxiliary potential $V_1(x)$ using the recurrence relation

$$\begin{aligned}V_1(x) &= -V_0(x) + 2E_0 + \frac{\hbar^2}{M}\left(\frac{\phi_0'(x)}{\phi_0(x)}\right)^2 \\ &= -\frac{1}{2}M\omega^2x^2 + \hbar\omega + \frac{\hbar^2}{M}(-cx)^2 \\ &= \frac{1}{2}M\omega^2x^2 + \hbar\omega = V_0(x) + \hbar\omega. \end{aligned} \tag{15}$$

Because the auxiliary potential $V_1(x)$ is identical to $V_0(x)$ up to a constant energy shift, the algebra simplifies significantly. The ground-state wavefunctions for all auxiliary Hamiltonians are identical to that of the original Hamiltonian and are given by $\phi_n(x) = \phi_0(x)$ for all n. This means the logarithmic derivative $\phi_n'(x)/\phi_n(x) = -cx$ is the same for all n in this chain. Furthermore, the second-order derivative term shared across the derivations simplifies to $\frac{2M}{\hbar^2}(V_n(x) - E_n) = c^2x^2 - c$.

Substituting these logarithmic derivatives and energy differences ($E_1 - E_0 = \hbar\omega$) into Eq. (11), we find

$$\psi_1(x) = \frac{-i\hbar}{\sqrt{2M(E_1-E_0)}}\left(\frac{\phi_1'(x)}{\phi_1(x)} + \frac{\phi_0'(x)}{\phi_0(x)}\right)\phi_0(x)$$
$$= \frac{-i\hbar}{\sqrt{2M\hbar\omega}}\left(-2cx\right)\phi_0(x) = i\sqrt{2c}x\phi_0(x). \qquad (16)$$

By factoring out the overall global phase factor of i, we recover the conventional formula for the first excited state, incorporating the first Hermite polynomial $H_1(\xi) = 2\xi$, where $\xi = \sqrt{c}x$.

Next, we evaluate the second excited state using Eq. (A12). Substituting the constant energy differences ($E_2 - E_1 = \hbar\omega$) and ($E_2 - E_0 = 2\hbar\omega$) yields

$$\psi_2(x) = \frac{-\hbar^2\phi_0(x)}{2M\sqrt{(E_2-E_1)(E_2-E_0)}}\left\{\left(\frac{\phi_2'(x)}{\phi_2(x)} + \frac{\phi_1'(x)}{\phi_1(x)}\right)\right.$$
$$\left.\times\left(\frac{\phi_1'(x)}{\phi_1(x)} + \frac{\phi_0'(x)}{\phi_0(x)}\right) + \frac{2M}{\hbar^2}\left(E_1 - E_2\right)\right\}$$
$$= \frac{-\hbar^2\phi_0(x)}{2M\sqrt{2(\hbar\omega)^2}}\left\{\left(-2cx\right)\left(-2cx\right) - 2c\right\}$$
$$= -\frac{1}{2\sqrt{2}}\left(4cx^2 - 2\right)\phi_0(x). \qquad (17)$$

Factoring out the global phase factor of -1, this also matches the conventional solution featuring the second Hermite polynomial $H_2(\xi) = 4\xi^2 - 2$.

Finally, we check the third excited state $\psi_3(x)$. The prefactor of the $\psi_3(x)$ equation becomes

$$\frac{(-i\hbar)^3}{\sqrt{(2M)^3(3\hbar\omega)(2\hbar\omega)(\hbar\omega)}} = \frac{i\hbar^3}{\sqrt{48M^3\hbar^3\omega^3}}$$
$$= \frac{i}{4\sqrt{3}}\left(\frac{\hbar}{M\omega}\right)^{3/2} = \frac{i}{4\sqrt{3}c^{3/2}}. \qquad (18)$$

Substituting the logarithmic derivative $-cx$ and the potential term $c^2x^2 - c$ into the expansion for $\psi_3(x)$ derived in Eq. (A16) simplifies the terms dramatically. The summation of the logarithmic derivative products yields $4c^3x^3$, while the potential-dependent terms evaluate to $-12c^3x^3 + 12c^2x$. Combining these, the term neatly collapses to $(-8c^3x^3 + 12c^2x)$. Multiplying this by the prefactor gives

$$\psi_3(x) = \frac{i}{4\sqrt{3}c^{3/2}}\left(-8c^3x^3 + 12c^2x\right)\phi_0(x)$$
$$= \frac{-i}{\sqrt{3}}\left(2c^{3/2}x^3 - 3c^{1/2}x\right)\phi_0(x)$$
$$= \frac{-i}{\sqrt{3}}\left(2\xi^3 - 3\xi\right)\phi_0(x). \qquad (19)$$

This is the correct third excited state, given by $\psi_3^{\text{exact}}(x) = (2^3 3!)^{-1/2}H_3(\xi)\phi_0(x)$. Since the third Hermite polynomial is $H_3(\xi) = 8\xi^3 - 12\xi$, the normalized state is $(48)^{-1/2}(8\xi^3 - 12\xi)\phi_0(x) = \frac{1}{\sqrt{3}}(2\xi^3 - 3\xi)\phi_0(x)$. Aside from the global phase factor of $-i$ the expression in the appendix yields the standard result.

While this served as a good check of the formulas, because the auxiliary wavefunctions were the same for each case, it collapsed a number of expressions to simpler forms. As a second check, we examine the simplified trigonometric Pöschl-Teller potential, given by

$$V_0(\hat{x}) = \frac{\hbar^2k^2\lambda(\lambda+1)}{2M}\sec^2(k\hat{x}), \qquad (20)$$

with k a wavenumber with dimensions of inverse length and λ a constant that determines the shape of the potential. This potential is valid in the domain $-\frac{\pi}{2k} \le x \le \frac{\pi}{2k}$. Furthermore, the logarithmic derivatives are no longer identical for this case, so it is a more complete check.

The exact ground-state energy and normalized wavefunction are

$$E_0 = \frac{\hbar^2k^2}{2M}(\lambda+1)^2, \quad \phi_0(x) = N_0\cos^{\lambda+1}(kx), \qquad (21)$$

where N_0 is the normalization constant. Because this potential is shape-invariant, the sequence of auxiliary Hamiltonians can be generated simply by shifting the shape parameter $\lambda \to \lambda + n$. Hence, the nth auxiliary potential, energy, and logarithmic derivative are determined by

$$V_n(x) = \frac{\hbar^2k^2}{2M}(\lambda+n)(\lambda+n+1)\sec^2(kx),$$
$$E_n = \frac{\hbar^2k^2}{2M}(\lambda+n+1)^2,$$
$$\frac{\phi_n'(x)}{\phi_n(x)} = -k(\lambda+n+1)\tan(kx). \qquad (22)$$

The nth auxiliary ground state has the same functional form as the ground-state does, but with $\lambda \to \lambda + n$ and a correction according to the normalization constant for that auxiliary ground state, given by N_n, which needs to be determined in the standard way. By substituting these parameter-shifted expressions into Eq. (11), we find

$$\psi_1(x) = i\sqrt{2\lambda+3}\tan(kx)\phi_1(x). \qquad (23)$$

Applying this exact same direct substitution protocol to Eq. (A12) yields the state

$$\psi_2(x) = -\frac{1}{2}\sqrt{\frac{2\lambda+5}{\lambda+2}}\left[(2\lambda+3)\tan^2(kx) - 1\right]\phi_2(x). \qquad (24)$$

Finally, the third excited state formula in Eq. (A16) produces

$$\psi_3(x) = i\sqrt{\frac{(2\lambda+5)(2\lambda+7)}{12(\lambda+3)}}$$
$$\times\left[3\tan(kx) - (2\lambda+3)\tan^3(kx)\right]\phi_3(x). \qquad (25)$$

We have checked that these are the first three excited states of the Pöschl-Teller potential, which verifies the formulas we use. We have suppressed the very long algebra involved in these derivations.

B. Numerical verification with shape invariant potentials

In order to perform numerics, we need to work with dimensionless variables. The energy will be written in terms of $\hbar^2 k^2/M$, with $X = kx$ being dimensionless. For the harmonic oscillator, the Hamiltonian is rewritten as

$$\hat{H}_{SHO} \to -\frac{\hbar^2}{2M}\frac{d^2}{dx^2} + \frac{1}{2}M\omega^2 x^2 = \hbar\omega\left(-\frac{1}{2}\frac{d^2}{dX^2} + \frac{1}{2}X^2\right) \tag{26}$$

in the position representation, with $k = \sqrt{M\omega/\hbar}$ the inverse length scale. Using this value of k, we have that $\hbar^2 k^2/M = \hbar\omega$.

We run the DVR solver with $-7 \le X \le 7$ and $N = 1401$ grid points with $-700 \le m,n \le 700$, so $\Delta X = k\Delta x = 0.01$. Then the Hamiltonian matrix becomes

$$\begin{aligned}\frac{H_{nn}}{\hbar\omega} &= \frac{5000\pi^2}{3} + 0.00005n^2 \;\text{ and}\\ \frac{H_{mn}}{\hbar\omega} &= \frac{10\,000(-1)^{m-n}}{(m-n)^2} \;\text{ for }\; m \ne n.\end{aligned} \tag{27}$$

Running our code gave a root mean square error between the numerical and analytic results of less than 10^{-9}, verifying the DVR wavefunction solver works well. We also ran the code to compute the excited energy eigenstates using the formulas for the ψ's in terms of the ϕ's. The continuous derivatives required to iteratively construct the sequence of auxiliary potentials and the final wavefunctions are approximated using second-order central finite differences, given by

$$\phi'(X_i) = \frac{\phi(X_{i+1}) - \phi(X_{i-1})}{2\Delta X}, \tag{28}$$

with one-sided differences used on the boundaries. The error in this numerical derivative propagates through all of the formulas. For example, for the third excited state, the numerical energy is equal to $E_3 = 3.500015\hbar\omega$, matching the exact analytical value of $3.5\hbar\omega$ to almost 6 significant digits. When comparing the wavefunctions, we find that the third excited state yields a root-mean-square error of 4.7×10^{-5} when compared to the exact analytic wavefunction. This deviation is consistent with the expected $O(\Delta X^2)$ truncation error from the finite difference derivatives. We can see, however, that the errors are larger than from directly diagonalizing the potentials. Eventually these errors will become significant causing increasing errors unless more accurate methods are used to calculate the numerical derivatives, or the step size is reduced. We do not pursue those advances here though, as they are not needed for our work.

For the trigonometric Pöschl-Teller potential, the Hamiltonian is rewritten in its dimensionless form as

$$\hat{H}_{PT} \to \varepsilon\left(-\frac{1}{2}\frac{d^2}{dX^2} + \frac{1}{2}\lambda(\lambda+1)\sec^2(X)\right) \tag{29}$$

in the position representation. Here, $X = kx$ and $\varepsilon = \hbar^2k^2/M$. Note that this potential is strictly defined within the domain $-\pi/2 < X < \pi/2$.

We run the DVR solver with $N = 315$ grid points defined by a symmetric index grid $-157 \le m,n \le 157$. Choosing a step size of $\Delta X = 0.01$, the grid spans $-1.57 \le X \le 1.57$, safely keeping the endpoints just inside the asymptotes at $\pm\pi/2 \approx \pm1.5708$. The dimensionless Hamiltonian matrix becomes

$$\begin{aligned}\frac{H_{nn}}{\varepsilon} &= \frac{5000\pi^2}{3} + 0.5\lambda(\lambda+1)\sec^2(0.01n) \;\text{ and}\\ \frac{H_{mn}}{\varepsilon} &= \frac{10\,000(-1)^{m-n}}{(m-n)^2} \;\text{ for }\; m \ne n.\end{aligned} \tag{30}$$

Because the DVR basis assumes the wavefunction is zero outside the defined grid, truncating the space just inside the asymptotes naturally enforces the required boundary conditions. We use the concrete choice of $\lambda = 3/2$ when we perform the numerical calculations. When we compare the DVR results to the analytical results, we find that the energies are exact and the RMS error in the wavefunctions become about 10^{-7} for the third excited state. Given the simple way we are handling the behavior of the divergent potential at the boundaries, these are good results.

To verify the numerical factorization chain, we extract the numerical auxiliary ground states sequentially and use our formulas to reconstruct the excited states. For the third excited state, the sequence yields a numerical energy of $E_3 = 15.122931\varepsilon$, matching the exact analytical value of 15.125ε ($\lambda = 3/2$). Reconstructing the third excited wavefunction via the numerical factorization method yields a root-mean-square error of 4.1×10^{-4} when compared to the exact result. This error is slightly larger than the harmonic oscillator case, which is a direct consequence of the spatial asymptotes of the potential at $X = \pm\pi/2$. The steep boundary potential forces sharp wavefunction curvature, moderately amplifying the $O(\Delta X^2)$ truncation error inherent to the discrete finite difference derivatives used to generate the successive auxiliary potentials. Nevertheless, the high degree of accuracy maintained across three generations of numerical differentiation confirms the validity of our approach.

C. Quartic potential

Next, we tackle the supersymmetry breaking case with

$$V_0(x) = \frac{\hbar^2k^6x^4}{2M} - \frac{\hbar^2k^3x}{M} = \frac{\hbar^2k^2}{M}\left(\frac{1}{2}X^4 - X\right). \tag{31}$$

The presence of the linear term breaks the even symmetry of a pure quartic well, distorting wavefunctions to have higher amplitudes on the lower potential side of the distortion. Because the asymptotic behavior is dominated by the X^4 term, solving this system numerically is significantly more demanding than the harmonic oscillator; it possesses complex, nonanalytic wavefunctions that decay as $\exp(-\frac{1}{3}|X|^3)$. This rapid decay introduces accuracy issues for numerical solvers when X becomes too large in magnitude, due to the sharp decrease of the exponential function.

To numerically construct the standard Schrödinger factorization chain, we utilize the DVR method described above.

The DVR method provides exceptional accuracy for all wavefunctions and energies where the wavefunction is essentially zero at the boundaries of the position domain included in the numerical calculation. The main numerical errors in the construction come primarily from the need to compute numerical logarithmic derivatives $\phi_n'(X)/\phi_n(X)$. In fact, when we are in the forbidden region, where X lies outside the auxiliary Hamiltonian turning points for the given n, the function $\phi_n(X)$ will have a small magnitude, so the accuracy of the logarithmic derivative becomes quite poor. For this system, we find the errors become large when $|X| \gtrapprox 3$. A simple analysis of the Schrödinger factorization chain shows that the asymptotic behavior of the superpotential must take the form

$$W_n(X) \approx \mathrm{sgn}(X)X^2 \tag{32}$$

for large enough $|X|$. We solve this numerical issue by using a regularized boundary matching technique, which employs a local quadratic extrapolation,

$$\phi_n'(X)/\phi_n(X) \approx a_n X^2 + b_n X + c_n, \tag{33}$$

fit to the DVR data just inside the precision floor determined by the X value on the left and on the right where $|\phi_n(X)| < 10^{-5}$. We need to fit with a different quadratic on the left and right sides. Our code performs the fit using the 20 points of data closest to but in the interior to the threshold X value, which always lies around $|X| \approx 3$. The leading term enforces the physically mandated parabolic envelope, while the linear and constant terms provide extra degrees of freedom required to guarantee continuity of the extrapolation with the numerically generated data. This approach is, in essence, an ansatz we tried, and as you can see from the accuracy we achieved, it works quite well. Without the extrapolation, the numerical calculations either need to be truncated within the $|X| \leq 3.5$ domain, or they will yield results that are dominated by numerical noise.

Table I. Comparison of the exact numerical energies of V_0 with the corresponding ground and excited states of the partner potentials V_1, V_2, and V_3 generated via the numerical factorization algorithm. Errors are in parentheses. Energies are in units of $\hbar^2 k^2/2M$.

State	Exact V_0	V_1 Energy (ΔE)	V_2 Energy (ΔE)	V_3 Energy (ΔE)
0	0.281068	–	–	–
1	1.854587	1.854531 (5.58×10^{-5})	–	–
2	3.686420	3.686331 (8.87×10^{-5})	3.686254 (1.65×10^{-4})	–
3	5.789275	5.789232 (4.26×10^{-5})	5.789091 (1.83×10^{-4})	5.788986 (2.88×10^{-4})

To verify the accuracy of this approach, we examine the lowest four energies for V_0, the lowest three energies for V_1, and so on, computed exactly with the DVR for the potentials that we found by numerically implementing the Schrödinger factorization chain. In the exact solution, the energies should be equal across the different chain elements. We find an error on the order of $4 \times 10^{-5} - 3 \times 10^{-4}$, as shown in Tab. I. The accuracy achieved by this approach confirms that the boundary regularization procedure that we used is a valid approach to handle the numerical precision required. We also examined the excited state wavefunctions of V_0 as calculated with our numerical chain, to the results found from the DVR calculations of the wavefunctions of V_0, which we view as the exact results. For this, we also found similar root-mean-square errors in the range from $10^{-5} - 10^{-4}$. In these calculations we used $X_{\max} = 7$ and $\Delta X = 0.01$ for a total of 1401 grid points in the calculations.

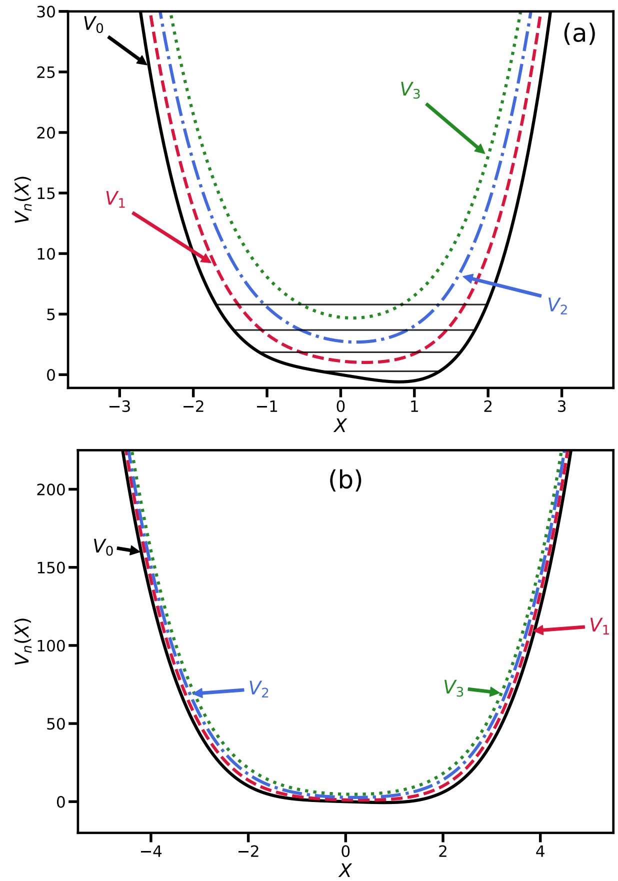

Figure 2. (a) Low energy region of the potentials found from the numerical factorization chain. The horizontal lines show the first four energy levels of V_0. (b) Higher-energy region of the potentials. By looking at the region near $|X| \approx 3$, we see that the potential is smooth there with no kinks, indicating the extrapolation procedure used is not introducing any artificial kinks into the potentials.

In Fig. 2, we plot the four potentials found with the numerical calculations (initial and first three auxiliary Hamiltonian potentials). In the figure, we also show the lowest energy levels for the different potentials as well. We find that the low-energy part of the potential becomes less asymmetric as we proceed down the chain. If we look at the shape of the potentials at $|X| \approx 3$, we see that the potentials are smooth there, further indicating that the boundary regularization procedure we use is accurate.

We run a second test of the accuracy of the approach by looking at all energies of the four potentials up to $E \approx 360\hbar^2k^2/M$ (77 excited states). At these largest energies, the classical turning points lie at ($|X| \approx 5.2$), lying well inside the region where we use the extrapolation procedure. Because the energies for each of these potentials should be identical in the factorization chain, differences let us understand the accuracy of the numerical procedure. Because the errors grow with n, the accuracy will be the worst for V_3. Despite the compounding numerical noise inherent in calculating V_3, the global RMS error for these highly excited states remains capped at roughly 0.9% over the 77 excited levels. This shows that the shape of the potentials determined by this numeric procedure remains good even to higher energy states, even though we see a clear degradation of the results for the higher energy levels.

In all cases, making the step size smaller will improve the accuracy of the numerics, but it will make the calculations take more compute time.

IV. CLASSROOM ADVICE

One of the main points of this work is to overcome the misconception in supersymmetric quantum mechanics that a specific type of broken supersymmetry is associated with a particular potential. The proper viewpoint is that broken supersymmetry is associated with the superpotential, and because factorizations need not be unique, a standard factorization chain can always be constructed even for systems with supersymmetry breaking. It is just that the partner potentials using the standard factorization are different (and do not break supersymmetry). In other words, supersymmetry breaking is always associated with the superpotential, not with the potential.

Much of this work can be transformed into an extended numerical project, especially if it is focused on more ways to numerically perform factorization. Our work here is just the tip of the iceberg for how one can work with numerical factorization of Hamiltonians. Particularly interesting would be a generalization of the numerics to be able to perform factorization without requiring solving the Schrödinger equation for the ground state of each auxiliary potential (this is an open-ended problem which has not been solved). Other projects can simply examine different types of initial potentials to see how the potentials evolve in the factorization chain or to examine isoenergetic potentials.

As we saw with this work, the numerics gets challenging when dealing with higher-order derivatives. Another interesting area to study is the extrapolation approach used for large $|x|$.

Finally, we note that one can have students work on this project using LLMs to generate the codes based on prompts that the students use. We used this technique in generating our codes, and found it was a very efficient process that allowed us to perform the analysis substantially faster than we would have had we written the codes ourselves; we used the Gemini 3.1 Pro model for all code generation. While this proposition may be controversial to some, we feel it has a number of benefits. Namely, it does not require the students to know how to code before they engage in this type of numerics. It allows students to focus on the issues we care about, such as accuracy, convergence, and physical interpretations of the numerical results. Instruction can then focus on providing strategies for useful prompts and in how to validate and verify the results of the computations.

V. SUMMARY AND CONCLUSIONS

In this work, we showed how one can numerically implement the first few steps of a factorization chain applied to exactly solvable cases (to check the accuracy of the approach) and a system that illustrates supersymmetry breaking (to clarify how supersymmetry breaking works). The numerical approach we applied is straightforward, but becomes increasingly difficult to carry out to arbitrary order. It would be interesting to see alternative numerical methods for factorization, especially ones that can develop the factorization without having to first solve a Schrödinger equation.

VI. ACKNOWLEDGMENTS

R. S. was supported by the National Science Foundation under Grant No. DMR-1950502. B.L. and J.K.F. were supported by the National Science Foundation under Grant No. CHE-2154152. J.K.F. was also supported by the McDevitt bequest at Georgetown University. J.K.F. came up with the idea for the project, which was examined in detail by B.L. and R.S. (L.X. acted as a mentor to B.L. and R.S. on the project). All authors contributed to the write-up of the work.

VII. AUTHOR DECLARATIONS

The authors have no conflicts to disclose.

Appendix A: Summary of formulas for higher excited states

Recall that the recurrence relation obtained from the factorization chain procedure for the potentials of arbitrary order is

$$V_{n+1}(x) = -V_n(x) + 2E_n + \frac{\hbar^2}{M}\left(\frac{\phi_n'(x)}{\phi_n(x)}\right)^2. \tag{A1}$$

The first three potentials are thus explicitly

$$V_1(x) = -V_0(x) + 2E_0 + \frac{\hbar^2}{M}\left(\frac{\phi_0'(x)}{\phi_0(x)}\right)^2, \tag{A2}$$

$$V_2(x) = V_0(x) - 2E_0 - \frac{\hbar^2}{M}\left(\frac{\phi_0'(x)}{\phi_0(x)}\right)^2 + 2E_1 + \frac{\hbar^2}{M}\left(\frac{\phi_1'(x)}{\phi_1(x)}\right)^2, \tag{A3}$$

$$V_3(x) = -V_0(x) + 2E_0 + \frac{\hbar^2}{M}\left(\frac{\phi_0'(x)}{\phi_0(x)}\right)^2 - 2E_1 - \frac{\hbar^2}{M}\left(\frac{\phi_1'(x)}{\phi_1(x)}\right)^2 + 2E_2 + \frac{\hbar^2}{M}\left(\frac{\phi_2'(x)}{\phi_2(x)}\right)^2. \tag{A4}$$

These potentials will be substituted into the derivation for higher excited states in the following. Next, let us recall the higher-order derivatives of the wavefunctions are given as follows

$$\phi_n''(x) = \frac{2M}{\hbar^2}(V_n(x) - E_n)\phi_n(x), \tag{A5}$$

$$\phi_n'''(x) = \frac{2M}{\hbar^2}(V_n'(x)\phi_n(x) + (V_n(x) - E_n)\phi_n'(x)). \tag{A6}$$

In the third-order derivative seen in Eq. (A6), a derivative of the potential needs to be calculated. For $n = 2$ and $n = 3$ that we will discuss in detail the derivatives of the potentials are

$$V_2'(x) = V_0'(x) + \frac{2\hbar^2}{M}\left(\frac{\phi_1'(x)}{\phi_1(x)}\frac{2M}{\hbar^2}(V_1(x) - E_1) - \left(\frac{\phi_1'(x)}{\phi_1(x)}\right)^3 - \frac{\phi_0'(x)}{\phi_0(x)}\frac{2M}{\hbar^2}(V_0(x) - E_0) + \left(\frac{\phi_0'(x)}{\phi_0(x)}\right)^3\right), \tag{A7}$$

$$V_3'(x) = -V_0' + \frac{2\hbar^2}{M}\left(\frac{\phi_0'(x)}{\phi_0(x)}\frac{2M}{\hbar^2}(V_0(x) - E_0) - \left(\frac{\phi_0'(x)}{\phi_0(x)}\right)^3 - \frac{\phi_1'(x)}{\phi_1(x)}\frac{2M}{\hbar^2}(V_1(x) - E_1) + \left(\frac{\phi_1'(x)}{\phi_1(x)}\right)^3 + \frac{\phi_2'(x)}{\phi_2(x)}\frac{2M}{\hbar^2}(V_2(x) - E_2) - \left(\frac{\phi_2'(x)}{\phi_2(x)}\right)^3\right). \tag{A8}$$

The second excited state is then given as

$$\begin{aligned}
|\psi_2\rangle &= \frac{1}{\sqrt{(E_2 - E_1)(E_2 - E_0)}}A_0^\dagger A_1^\dagger|\phi_2\rangle = \frac{1}{2M\sqrt{(E_2 - E_1)(E_2 - E_0)}}\left(p - i\hbar\frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right)\left(p - i\hbar\frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right)|\phi_2\rangle \\
&= \frac{-i\hbar}{2M\sqrt{(E_2 - E_1)(E_2 - E_0)}}\left(p - i\hbar\frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right)\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right)|\phi_2\rangle \\
&= \frac{(-i\hbar)^2}{2M\sqrt{(E_2 - E_1)(E_2 - E_0)}}\left\{\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right)\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right) + \frac{1}{-i\hbar}\left[p, \frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right]\right\}|\phi_2\rangle.
\end{aligned} \tag{A9}$$

The last step of the equations above is obtained by moving the term $(p - i\hbar\frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})})$ through the term $(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})})$, creating the commutator $[\hat{p}, \frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}]$. Such a step is necessary because we need the momentum operator $\hat{p}$ to act directly on the state $|\phi_n\rangle$ to be able to use the subsidiary condition in Eq. (9)

$$\hat{p}|\phi_n\rangle = -i\hbar\frac{\phi_n'(\hat{x})}{\phi_n(\hat{x})}|\phi_n\rangle. \tag{A10}$$

We will use this technique on many occasions throughout this section. The derivation for the second excited state follows as

$$\begin{aligned}
|\psi_2\rangle &= \frac{(-i\hbar)^2}{2M\sqrt{(E_2 - E_1)(E_2 - E_0)}}\left\{\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right)\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right) + \left(\frac{\phi_2''(\hat{x})}{\phi_2(\hat{x})} - (\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})})^2 + \frac{\phi_1''(\hat{x})}{\phi_1(\hat{x})} - (\frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})})^2\right)\right\}|\phi_2\rangle \\
&= \frac{(-i\hbar)^2}{2M\sqrt{(E_2 - E_1)(E_2 - E_0)}}\left\{\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right)\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right) + \frac{2M}{\hbar^2}(V_2 - E_2) - \left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})}\right)^2 \right. \\
&\qquad \left. + \frac{2M}{\hbar^2}(V_1 - E_1) - \left(\frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right)^2\right\}|\phi_2\rangle \\
&= \frac{(-i\hbar)^2}{2M\sqrt{(E_2 - E_1)(E_2 - E_0)}}\left\{\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})} + \frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right)\left(\frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})} + \frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right) + \frac{2M}{\hbar^2}(E_1 - E_2)\right\}|\phi_2\rangle.
\end{aligned} \tag{A11}$$

The second-order logarithmic derivative seen in the first line of the above derivation is substituted out by the explicit expression seen in Eq. (A5). The wavefunction is therefore

$$\psi_2(x) = \frac{(-i\hbar)^2\phi_2(x)}{2M\sqrt{(E_2-E_1)(E_2-E_0)}}\left\{\left(\frac{\phi_2'(x)}{\phi_2(x)}+\frac{\phi_1'(x)}{\phi_1(x)}\right)\left(\frac{\phi_1'(x)}{\phi_1(x)}+\frac{\phi_0'(x)}{\phi_0(x)}\right)+\frac{2M}{\hbar^2}(E_1-E_2)\right\}, \tag{A12}$$

So far we have only used second-order logarithmic derivatives of the superpotentials. For the third-order wavefunction, we need the third-order derivatives. Following Eq. (A6), we have

$$\begin{aligned}\phi_2'''(x) = \frac{2M}{\hbar^2}\Bigg\{\Bigg[V_0'(x)+\frac{2\hbar^2}{M}\Bigg(\frac{\phi_1'(x)}{\phi_1(x)}\frac{2M}{\hbar^2}(V_1(x)-E_1)-\left(\frac{\phi_1'(x)}{\phi_1(x)}\right)^3-\frac{\phi_0'(x)}{\phi_0(x)}\frac{2M}{\hbar^2}(V_0(x)-E_0)+\left(\frac{\phi_0'(x)}{\phi_0(x)}\right)^3\Bigg)\Bigg]\phi_2(x)\\ +(V_2(x)-E_2)\phi_2'(x)\Bigg\},\end{aligned} \tag{A13}$$

$$\begin{aligned}\phi_3'''(x) = \frac{2M}{\hbar^2}\Bigg\{\Bigg[-V_0'(x)+\frac{2\hbar^2}{M}\Bigg(\frac{\phi_0'(x)}{\phi_0(x)}\frac{2M}{\hbar^2}(V_0(x)-E_0)-\left(\frac{\phi_0'(x)}{\phi_0(x)}\right)^3-\frac{\phi_1'(x)}{\phi_1(x)}\frac{2M}{\hbar^2}(V_1(x)-E_1)+\left(\frac{\phi_1'(x)}{\phi_1(x)}\right)^3\\ +\frac{\phi_2'(x)}{\phi_2(x)}\frac{2M}{\hbar^2}(V_2(x)-E_2)-\left(\frac{\phi_2'(x)}{\phi_2(x)}\right)^3\Bigg)\Bigg]\phi_3(x)+(V_3(x)-E_3)\phi_3'(x)\Bigg\},\end{aligned} \tag{A14}$$

after we substitute $V_2'(x)$ and $V_3'(x)$ with Eqs. (A7) and (A8). The third excited state can be constructed by a chain of ladder operators acting on the ground state

$$\begin{aligned}|\psi_3\rangle &= \frac{1}{\sqrt{(E_3-E_2)(E_3-E_1)(E_3-E_0)}}\hat{A}_0^\dagger\hat{A}_1^\dagger\hat{A}_2^\dagger|\phi_3\rangle\\ &= \frac{1}{\sqrt{(2M)^3(E_3-E_2)(E_3-E_1)(E_3-E_0)}}\left(\hat{p}-i\hbar\frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right)\left(\hat{p}-i\hbar\frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right)\left(\hat{p}-i\hbar\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})}\right)|\phi_3\rangle\\ &= \frac{(-i\hbar)^3}{\sqrt{(2M)^3(E_3-E_2)(E_3-E_1)(E_3-E_0)}}\Bigg\{\left(\frac{\phi_3'(\hat{x})}{\phi_3(\hat{x})}+\frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right)\left(\frac{\phi_3'(\hat{x})}{\phi_3(\hat{x})}+\frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right)\left(\frac{\phi_3'(\hat{x})}{\phi_3(\hat{x})}+\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})}\right)\\ &+\left(\frac{\phi_3'(\hat{x})}{\phi_3(\hat{x})}+\frac{\phi_0'(\hat{x})}{\phi_0(\hat{x})}\right)\left(\frac{\phi_3''(\hat{x})}{\phi_3(\hat{x})}-\left(\frac{\phi_3'(\hat{x})}{\phi_3(\hat{x})}\right)^2+\frac{\phi_2''(\hat{x})}{\phi_2(\hat{x})}-\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})}\right)^2\right)\\ &+\left(\frac{\phi_3'(\hat{x})}{\phi_3(\hat{x})}+\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})}\right)\left(\frac{\phi_3''(\hat{x})}{\phi_3(\hat{x})}-\left(\frac{\phi_3'(\hat{x})}{\phi_3(\hat{x})}\right)^2+\frac{\phi_1''(\hat{x})}{\phi_1(\hat{x})}-\left(\frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right)^2\right)\\ &+\left(\frac{\phi_3'(\hat{x})}{\phi_3(\hat{x})}+\frac{\phi_1'(\hat{x})}{\phi_1(\hat{x})}\right)\left(\frac{\phi_3''(\hat{x})}{\phi_3(\hat{x})}-\left(\frac{\phi_3'(\hat{x})}{\phi_3(\hat{x})}\right)^2+\frac{\phi_2''(\hat{x})}{\phi_2(\hat{x})}-\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})}\right)^2\right)\\ &+\frac{\phi_3'''(\hat{x})}{\phi_3(\hat{x})}-3\frac{\phi_3'(\hat{x})\phi_3''(\hat{x})}{\phi_3^2(\hat{x})}+\frac{\phi_2'''(\hat{x})}{\phi_2(\hat{x})}-3\frac{\phi_2'(\hat{x})\phi_2''(\hat{x})}{\phi_2^2(\hat{x})}+2\left(\left(\frac{\phi_3'(\hat{x})}{\phi_3(\hat{x})}\right)^3+\left(\frac{\phi_2'(\hat{x})}{\phi_2(\hat{x})}\right)^3\right)\Bigg\}|\phi_3\rangle.\end{aligned} \tag{A15}$$

By projecting onto the position basis $\langle x|$, we map the operators to coordinates and substitute the higher-order derivatives of the superpotentials using Eqs. (A13) and (A14). This yields a final functional form given by

$$
\begin{aligned}
\psi_3(x) &= \langle x|\psi_3\rangle \\
&= \frac{(-i\hbar)^3\phi_3(x)}{\sqrt{(2M)^3(E_3-E_2)(E_3-E_1)(E_3-E_0)}}\Bigg\{\frac{\phi_3'(x)}{\phi_3(x)}\frac{\phi_2'(x)}{\phi_2(x)}\frac{\phi_1'(x)}{\phi_1(x)}+\frac{\phi_3'(x)}{\phi_3(x)}\frac{\phi_2'(x)}{\phi_2(x)}\frac{\phi_0'(x)}{\phi_0(x)}+\frac{\phi_3'(x)}{\phi_3(x)}\frac{\phi_1'(x)}{\phi_1(x)}\frac{\phi_0'(x)}{\phi_0(x)} \\
&+\frac{\phi_2'(x)}{\phi_2(x)}\frac{\phi_1'(x)}{\phi_1(x)}\frac{\phi_0'(x)}{\phi_0(x)}-2\frac{\phi_3'(x)}{\phi_3(x)}\left(\frac{\phi_2'(x)}{\phi_2(x)}\right)^2-\frac{\phi_0'(x)}{\phi_0(x)}\left(\frac{\phi_2'(x)}{\phi_2(x)}\right)^2-\frac{\phi_1'(x)}{\phi_1(x)}\left(\frac{\phi_2'(x)}{\phi_2(x)}\right)^2 \\
&-\frac{\phi_3'(x)}{\phi_3(x)}\left(\frac{\phi_1'(x)}{\phi_1(x)}\right)^2-\frac{\phi_2'(x)}{\phi_2(x)}\left(\frac{\phi_1'(x)}{\phi_1(x)}\right)^2-2\left(\frac{\phi_2'(x)}{\phi_2(x)}\right)^3 \\
&+\left(\frac{\phi_3'(x)}{\phi_3(x)}+\frac{\phi_2'(x)}{\phi_2(x)}+\frac{\phi_1'(x)}{\phi_1(x)}+\frac{\phi_0'(x)}{\phi_0(x)}\right)\left(\frac{2M}{\hbar^2}\left(V_3(x)-E_3\right)\right) \\
&+\left(2\frac{\phi_3'(x)}{\phi_3(x)}+2\frac{\phi_2'(x)}{\phi_2(x)}+\frac{\phi_1'(x)}{\phi_1(x)}+\frac{\phi_0'(x)}{\phi_0(x)}\right)\left(\frac{2M}{\hbar^2}\left(V_2(x)-E_2\right)\right)+\left(\frac{\phi_3'(x)}{\phi_3(x)}+\frac{\phi_2'(x)}{\phi_2(x)}\right)\left(\frac{2M}{\hbar^2}\left(V_1(x)-E_1\right)\right)\Bigg\}.
\end{aligned}
\tag{A16}
$$

after we moved the term $(\hat{p}-i\hbar(\frac{\phi_n'(\hat{x})}{\phi_n(\hat{x})})$ through terms to utilize the subsidiary condition, creating commutators followed by substituting the third-order derivatives of the superpotentials with Eqs. (A13) and (A14). This final form is stable for the DVR method used for this work and it does not contain any derivatives higher than first-order derivatives.

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